

Autonomous Shopping Cart

PDE4435 - ASHWIN MURALI THANANPADI

APR - 2026

https://github.com/cruisemaniac/Smart_Cart_With_ROS_PDE4435_CW2

24-MAR

Git Repo Created

Basic repo structure is in place.

Need to work on responsibilities

I will work on Sensor Integration

27-MAR

Study UWB Sensors.

Understand how we decided on sensors for CW1

How will we get the sensors working in ROS2.

Read ROS2 !!!

29-MAR

Making for project structure & Git repo public access for everyone.

30-MAR

Ros2 Jazzy + **humble???** + Gazebo

03-APR

Vignesh will provide URDF.

Till then Jay & I will work on other aspects of code

09-APR

VWB Publish.

Something works. No clue how to fix this.
Need to ask someone how to do this.

11-APR

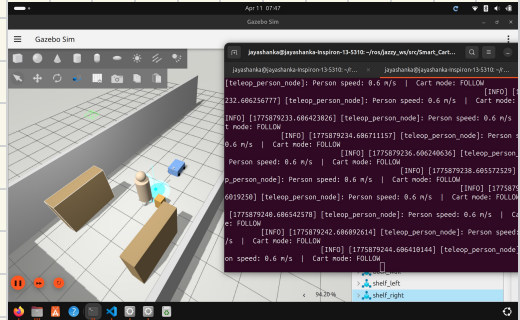
LIDAR or VWB discussion.

We like the multi sensor idea. Works for better accuracy but more complex.

What gives? **Can try to do this!?**

12-APRIL

VW3 simulation works.
Connected to /nub1distances
Person is at /person/odom

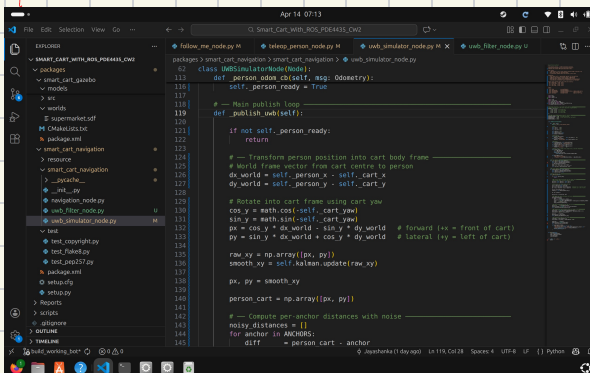


13-APRIL

VW3 Stable

15-APRIL

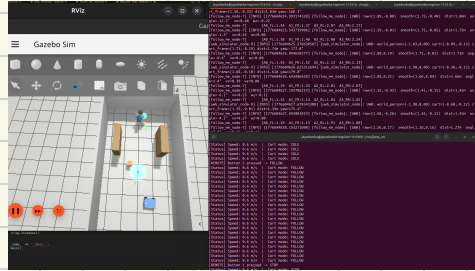
Least Squares Trilateration. Got x, y from
4 ranges
Kalman filter added. Needs tuning



19-APR

Ros2 Gazebo bridge works!!!

Full launch works



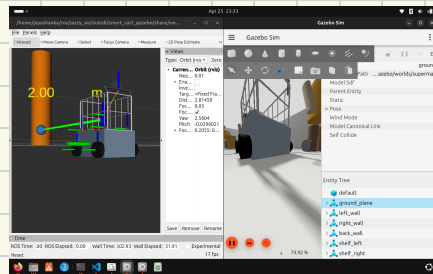
Sensor flow stable and works.
Can get data from gz sim.

23-APR

Fixed launch issues
Testing sensor nodes in sim.
Follow mode started - yay hxxx!!!
Testing followme mode to make sure
it works.

27-APR

Done Done!



All Sim based objectives achieved